Modeling neural networks with hypergraphs via tensor-based views

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Description of neural network (HyMeKo)

```
[ mlp_example import "meta_neural.himeko" ]
mlpminimal use meta.elements {
   input_layer: |> -> linear {
      in_features 4
      out_features 16
   }
   hidden_layer: |> -> linear {
      in_features 16
      out_features 8
   }
   output_layer: |> -> linear {
      in_features 8
      out_features 3
   }

   @connections {
      @e1: activations.relu {<- input_layer, -> hidden_layer},
      @e2: activations.relu {<- hidden_layer, -> output_layer},
      @e3: activations.softmax {<- output_layer}
   }
}
```

Difference in entropy using Kullbeck-Leibler divergence:

$$\mathcal{D}_{KL}(H_1, H_2) = -\sum_i H_1 \circ \log\left(\frac{1}{\hat{L}(H_2)}\right)$$

Normalized Laplacian

$$\hat{L}(H) = \frac{L(H)}{\sum_{x\in V} D(x)} = \frac{D(H) - A(H)}{\sum_{x\in V} D(x)}$$

Eigenvalues: $(\hat{\lambda}_i)_{i\in\{1,2,\dots,n\}}$

Algebraic entropy: $I(H) = -\sum_{i=1}^{n} \hat{\lambda}_i \log_2 \hat{\lambda}_i$

Cross entropy: $I_H(H_1, H_2) = -\sum_i \hat{L}(H_1) \circ \log \hat{L}(H_2)$

Add entropy difference to the total cost function

Training

Total cost function:

$$J(\theta_{t+1}) = J(\theta_t) + otherparts + \mathcal{D}_{KL}(H(\theta_t), H(\theta_{t+1}))$$

Transform to hypergraph description

Algebraic entropy of normalized tensor Laplacian

START

Neural network meta description

Graph elements describing a neural network (e.g., layer types, activations)

Runnable Neural Network code

Hypergraph-based description (HyMeKo)

HyMeKo framework

Hypergraph Text Transformation

Initial tensor view of hypergraph

Convert AST to processable hypergraphs

Parser

HyMeKo AST

AST-to-model transformer

HyMeKo intermediate model

Hypergraph tensor transformation

Parse the input text description

Represents the language model in a tree structure

(Dir)Hypergraph repr.: H_1, H_2

Fill tensors based on clique- or star-expansion

Resnet

Original notion of architecture

Transformation - expansion

Factor view

Dataflow view

Expanded view of neural network archtectures

References

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